

Interview Question :

U, V are standard normal, $\text{Corr}(U, V) = -0.72$

Find $E(3U + V \mid U - V = 1)$

$$Y = 3U + V, \quad X = U - V, \quad E(Y \mid X = 1)$$

If (X, Y) is bivariate normal, then

$$\begin{aligned} E(Y \mid X = x) &= \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \\ &= \mu_Y + \frac{\text{Cov}(X, Y)}{\sigma_X^2} (x - \mu_X) \end{aligned}$$

$$\text{Cov}(aX + bY, cU + dV) = ac \text{Cov}(X, U) + ad \text{Cov}(X, V) + bc \text{Cov}(Y, U) + bd \text{Cov}(Y, V)$$

$$\text{Cov}(X, Y) = V(X), \quad \text{so} \quad E(Y \mid X = 1) = 1$$

$$U = V + 1$$

$$E(X) = E(E(X \mid Y))$$

1 Heteroskedasticity

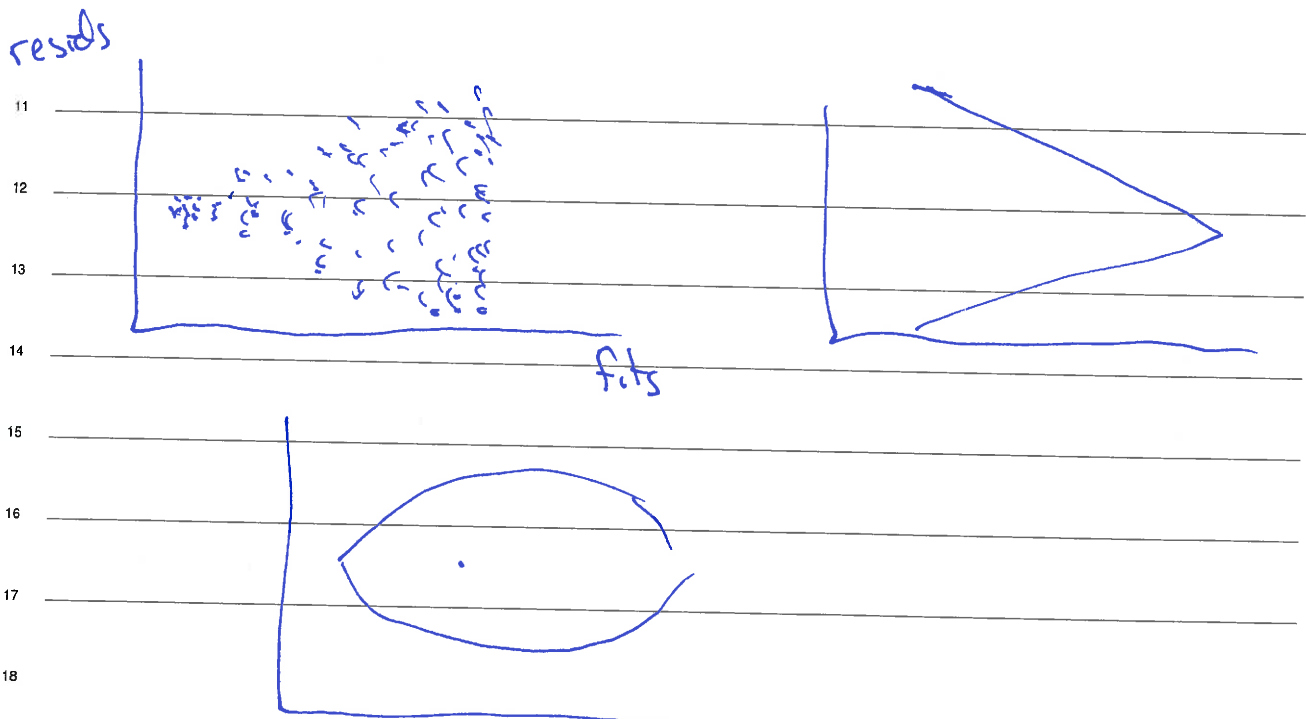
2 One assumption of our model is that all of the model errors have the same
3 variance; this condition is called **homoskedasticity**.

4 The opposite case, the presence of **heteroskedasticity** is a serious concern.

5 The validity of the confidence intervals, hypothesis tests, and so forth rests
6 on this assumption.

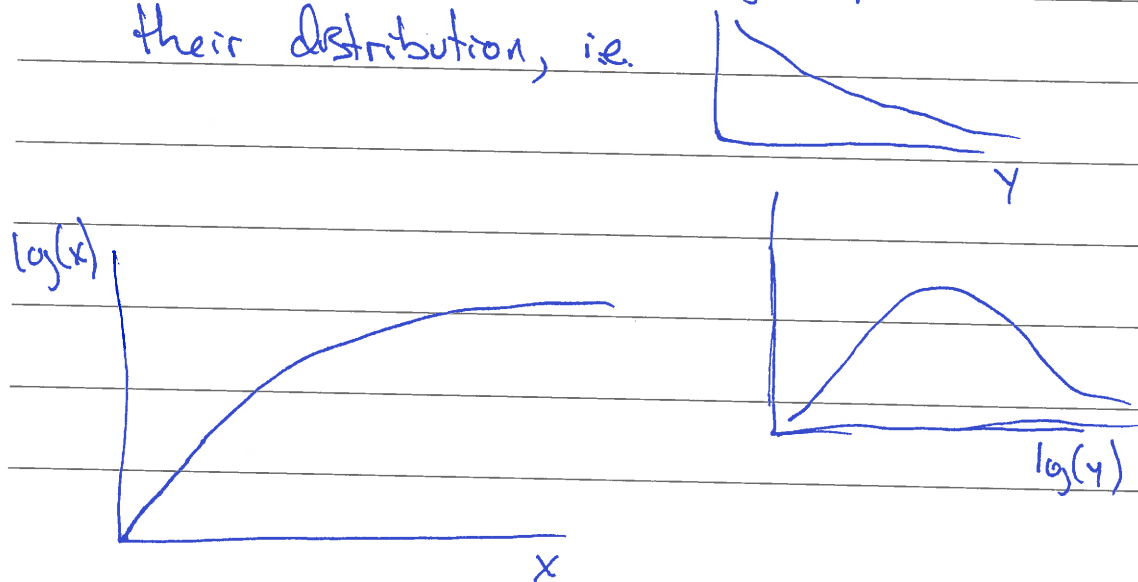
7 The plot of residuals versus fitted values is a useful tool for diagnosing
8 heteroskedasticity.

9 **Exercise:** What shapes would one expect to see in the plot of residuals
10 versus fitted values when heteroskedasticity is present?



- 1 There are two major strategies for dealing with heteroskedasticity.
- 2 First, the response variable can be **transformed**, meaning you would use
- 3 $g(y)$ instead of y as the response variable. The two most popular choices
- 4 for $g()$ are the logarithm, and the square root.
- 5 **Exercise:** Why do you think that the log transform is so often used with
- 6 financial variables (prices, volume, etc.)?

Many such variables have long upper tails to their distribution, i.e.



large, extreme values tend to have larger variance in their errors

- 1 The other major strategy for dealing with heteroskedasticity is to use **weighted**
 2 **least squares**. Briefly stated, instead of minimizing the residual sum of
 3 squares, one minimizes

$$4 \quad \text{WRSS}(\beta) = \sum_{i=1}^n w_i (y_i - \hat{y}_i)^2, \quad w_i \geq 0$$

- 5 where the weights are chosen to deemphasize those observations for which
 6 the variance is larger.

- 7 In fact, the optimal choice is $w_i = 1/\sigma_i^2$, where σ_i^2 is the variance for the i^{th}
 8 model error.

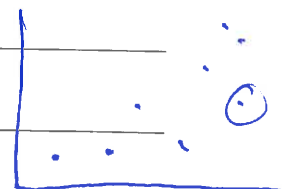
- 9 One would not typically know the σ_i^2 , but it is sometimes the case that you
 10 are willing to assume that $\sigma_i^2 = \sigma^2/m_i$. Then, the weight can be chosen as
 11 $w_i = m_i$. Or, the σ_i^2 could be estimated from the residuals.

- 12 **Exercise:** Explain why it may be the case that you would be willing to
 13 assume that $\sigma_i^2 = \sigma^2/m_i$.

14 Sometimes, the response y_i is actually the sample
 15 average of m_i observations which can be
 16 taken to have the same variance σ^2

$$17 \quad V(y_i) = \sigma^2/m_i$$

18 $\nwarrow$ sample size in group i



1 Example

2 In this example we will use 82 NYSE stocks in an effort to predict the real-
3 ized volatility, here defined as

$$4 \quad \frac{1}{n} \sum_{i=1}^m \left(\log \left(\frac{P_i}{P_{i-1}} \right) \right)^2 \quad m = \# \text{ of days}$$

5 where P_i is the closing price on day i .

6 For this simple model, we will use the logarithm of the total trading vol-
7 ume over the prior month as the predictor of the realized volatility over
8 the upcoming month.

9 Of course, to **train** or **fit** the model we need data on both the predictor and
10 the response. So, we use two recent consecutive months to form a training
11 set. We will use the aforementioned set of 82 stocks for this purpose.

12 A simple linear model is fit in attempt to model the relationship between
13 these two variables. Figure 12 shows the raw data and the result of the fit.

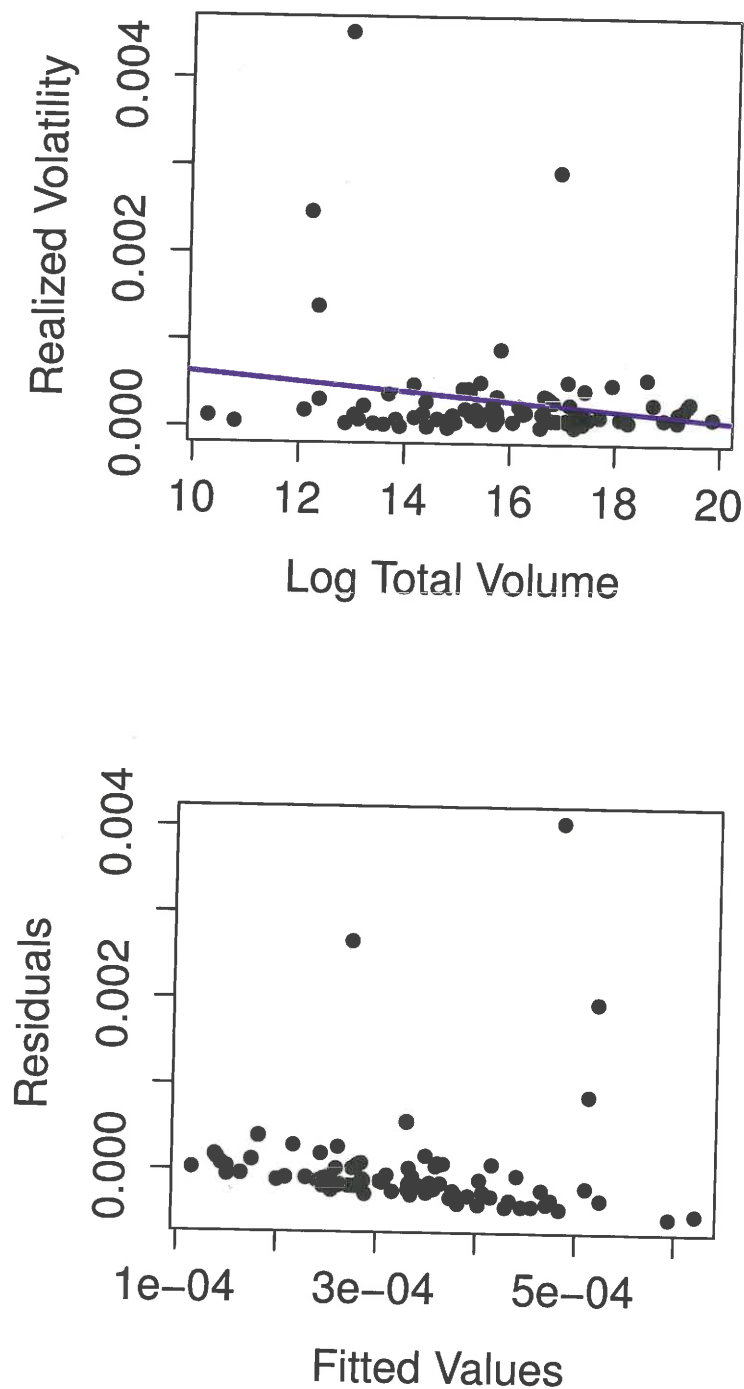


Figure 12: The initial model fit to realized volatility. The upper plot shows the scatter plot with the regression line superimposed, the lower plot shows the plot of residuals versus fitted values.

1 **Exercise:** Comment on the quality of the fit of this first model.

2 Evidence of lack of fit from the plot of
3 residuals versus fitted values. (Don't see
4 random scatter.)
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18 The model was refit with the four extreme observations removed. Figure
19 13 shows the result.

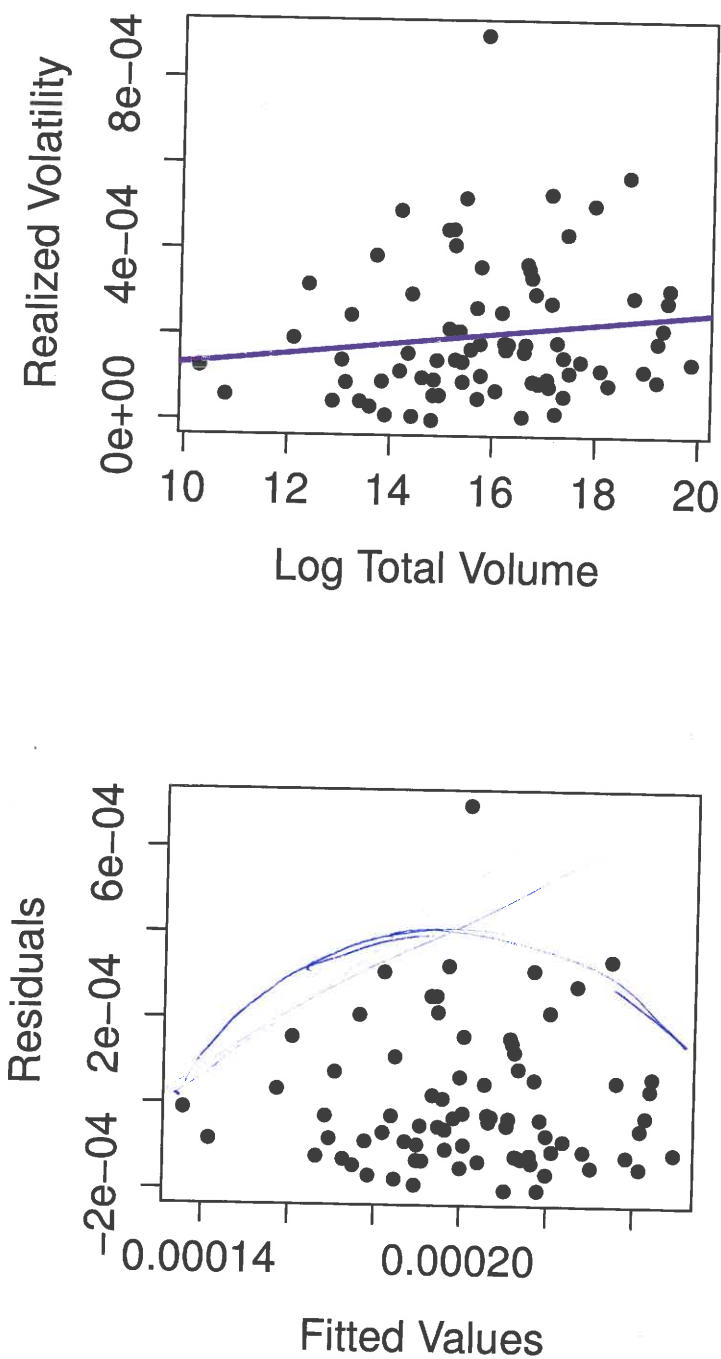


Figure 13: The model fit to realized volatility after the four extreme observations have been removed. The upper plot shows the scatter plot with the regression line superimposed, the lower plot shows the plot of residuals versus fitted values.

1 **Exercise:** Comment on the quality of the fit of this second model.

2 mild evidence of heteroskedasticity

18 The model was then refit with the four extreme observations returned, but
19 using the log transform of the response. Figure 14 shows the result.

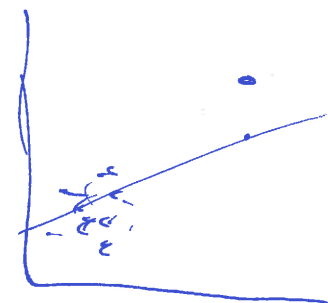
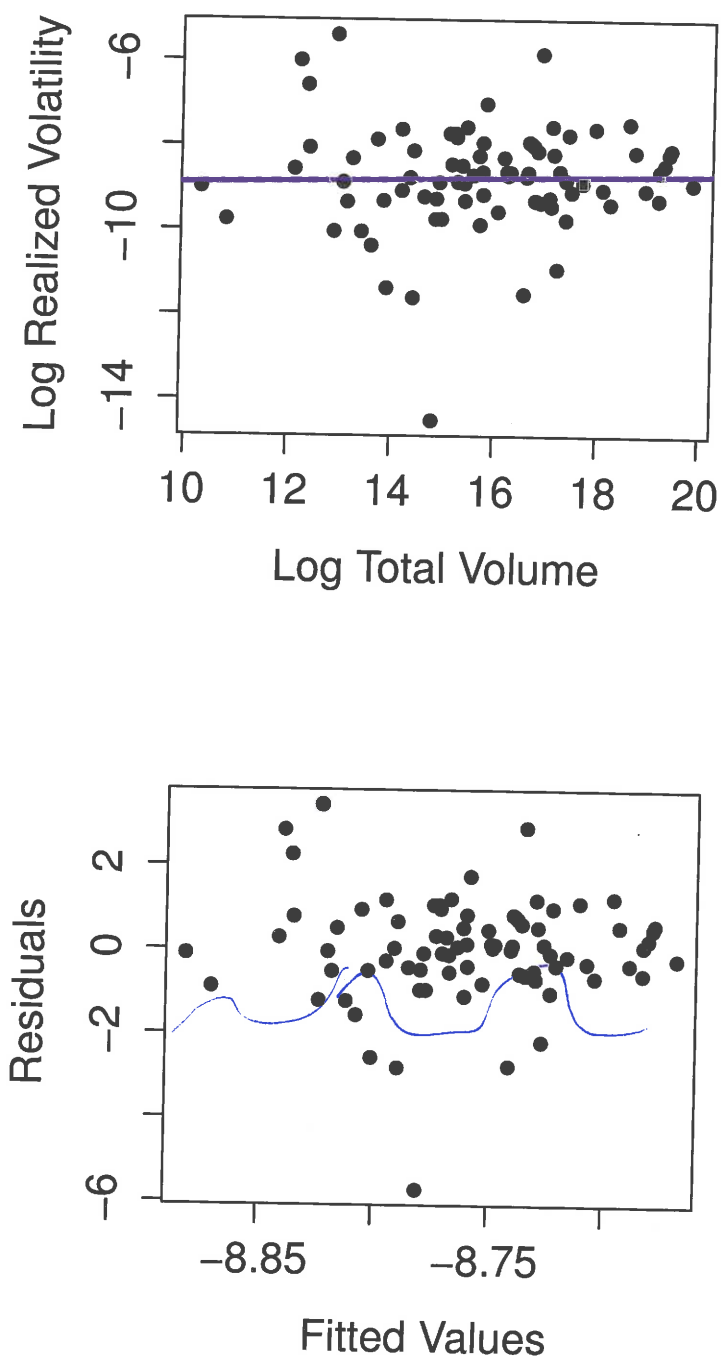


Figure 14: The model fit to realized volatility after taking the log transform. The upper plot shows the scatter plot with the regression line superimposed, the lower plot shows the plot of residuals versus fitted values.

1 **Exercise:** Comment on the quality of the fit of this third model.

2 Seems to have improved the quality of the fit.

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4 There is evidence of an "outlier" which should
5 be studied further.

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1 The Normality Assumption

2 We should also assess the validity of the assumption that the model errors
3 are normally distributed. This is typically done by looking at a normal
4 probability plot of the residuals.

5 The distribution of the $\hat{\beta}_i$ is only exactly normal when the model errors are
6 normal. But, there is a version of the central limit theorem that states that
7 $\hat{\beta}_i$ will be approximately normal as long as the sample size is sufficiently
8 large.

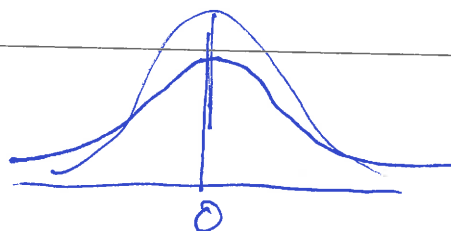
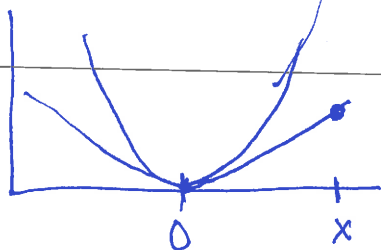
9 Hence, the standard errors, confidence intervals, and hypothesis tests we
10 construct will still be approximately valid if the model errors are not nor-
11 mal, as long as the sample size is sufficiently large.

1 Still, the normal probability plot is a valuable tool for identifying **outliers**,
2 which are objects with errors which are much larger than would be ex-
3 pected under the normal distribution.

4 Also, if we find that the error distribution has heavier tails than would be
5 expected under the normal distribution, we may want to reconsider our
6 use of least squares as the criterion for estimating β .

7 ★ **Exercise:** Explain why least squares may no longer be a good choice if
8 the distribution of the errors was, for instance, the t -distribution with 5
9 degrees of freedom.

10 The t -dist. with a few d.f. (< 10) has great potential
11 to generate large errors, obs. that fall far from
12 the regression line. Such obs. can have undue
13 influence on the estimated regression line when
14 using least squares. A penalty such as absolute
15 error is much more robust to extreme values



18 Later we will discuss **robust regression** approaches which use criterion
19 other than least squares.

$$rt(100, 1)$$

- 1 Figure 15 shows the normal probability plot for the example presented
- 2 above. This is for the case of the third model (Figure 14).

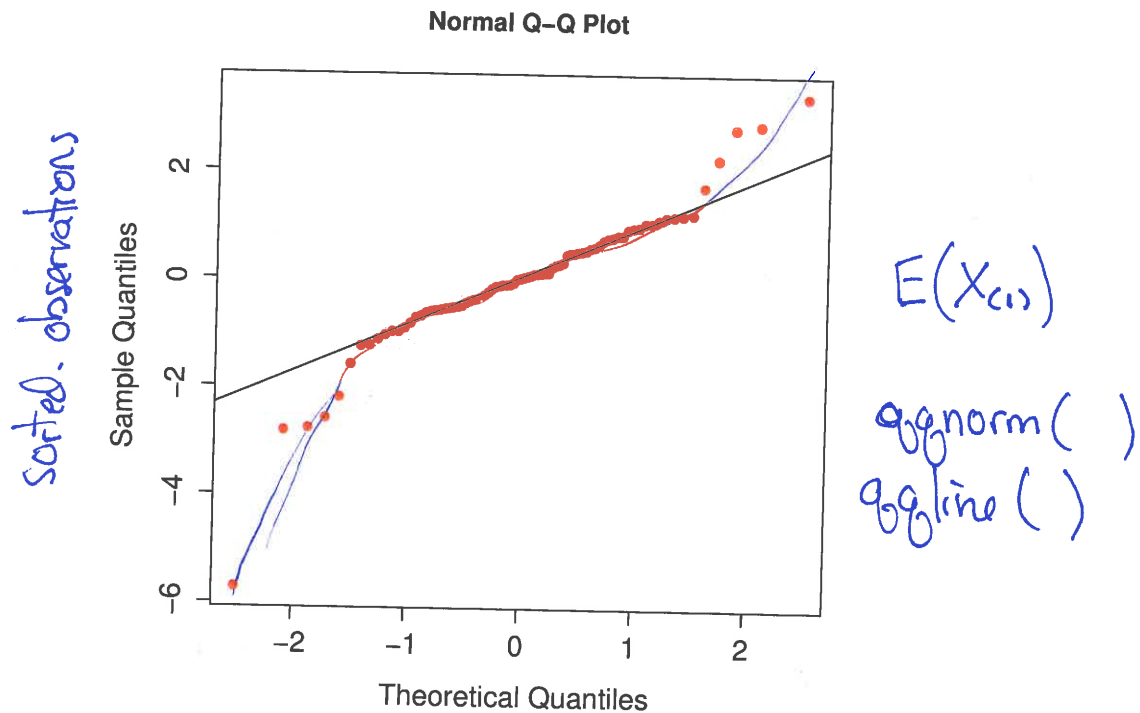


Figure 15: Normal probability plot corresponding to the model shown in Figure 14.

- 3 **Exercise:** Comment on the normal probability plot shown in Figure 15. Is
- 4 there an issue with outliers? Is the normal assumption valid?

Mild evidence that the dist. of the errors has heavier tails than the normal. There is one extreme value that is probably best considered an outlier. Overall, the normal assumption seems decent approx.

1 Other Diagnostic Plots

- 2 If there is a time dimension to the data that is being used (usually the case
- 3 with financial data), then one should plot the residuals versus time, and
- 4 see if there is time-dependence that is not being adequately modelled.
- 5 In a subsequent course you will learn about ARMA and GARCH models;
- 6 these can be built into the assumptions regarding the model error.
- 7 It is also a good idea to individually plot the residuals versus each of the
- 8 predictors in the model. If our model is adequate, there should be no rela-
- 9 tionship between these quantities.

Making Predictions

One of the fundamental motivations for using regression models is to make predictions for **future** observations of the response. In other words, consider a situation where you know the value of the predictor variable x , and seek to predict the response.

In the context of our simple linear regression model, if our value for the predictor is x^* , then we would start by finding

$$\hat{y}^* = \hat{\beta}_0 + \hat{\beta}_1 x^*$$

We have plugged in this new value for x into the line that has been fit. This is our best estimator for y^* , the true response for this object.

One should not be satisfied with only this, however: We should quantify the amount of error in this prediction.

The variance in the error in the prediction is

$$V(\hat{y}^* - y^*) = \sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2} \right]$$

↑
model
prediction
↑
actual
response

Exercise: Explain how the above decomposes the error in the prediction into two key sources.

$$V(\hat{y}^* - y^*) = \underbrace{\sigma^2}_{\text{irreducible error}} + \underbrace{\sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2} \right]}_{\text{error due to } \hat{\beta}_1 \neq \beta_1 \text{ and } \hat{\beta}_0 \neq \beta_0}$$

Note that this ignores error due to model misspecification

Hence the standard error in the prediction is

$$SE(\hat{y}^*) = \sqrt{\sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2} \right]}$$

Of course, in practice σ^2 will be replaced with $\hat{\sigma}^2$.

- 1 **Exercise:** What happens to the standard error in the prediction as n in-
2 creases?

3 As n increases, the variance of $\hat{\beta}_0$ and $\hat{\beta}_1$
4 converge to 0, hence the SE of the
5 prediction will be dominated by the
6 contribution from irreducible error.

7 The contrib. from irreducible error does
8 not vanish.
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- 16 A $100(1 - \alpha)\%$ **prediction interval for the future response** can be written
17 as

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$$\hat{y}^* \pm t_{\alpha/2, n-2} \text{SE}(\hat{y}^*)$$

- 19 In practice, we will rely on R to construct these intervals.

- 1 Figure 16 shows a simple example of how the prediction interval will ap-
- 2 pear, as x^* ranges over the extent of the predictor space.

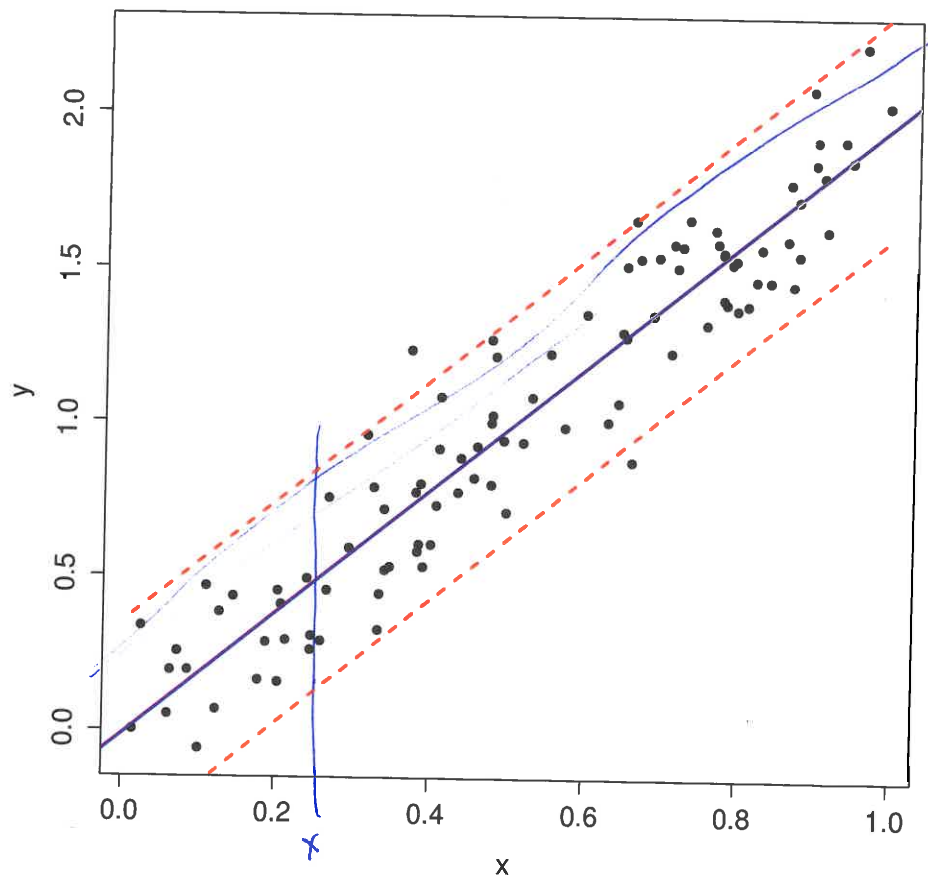


Figure 16: An example showing a regression line (solid line), along with the prediction bands (dashed lines).

Adding a Predictor

Practical regression models typically contain many predictors, i.e., we fit models of the form

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

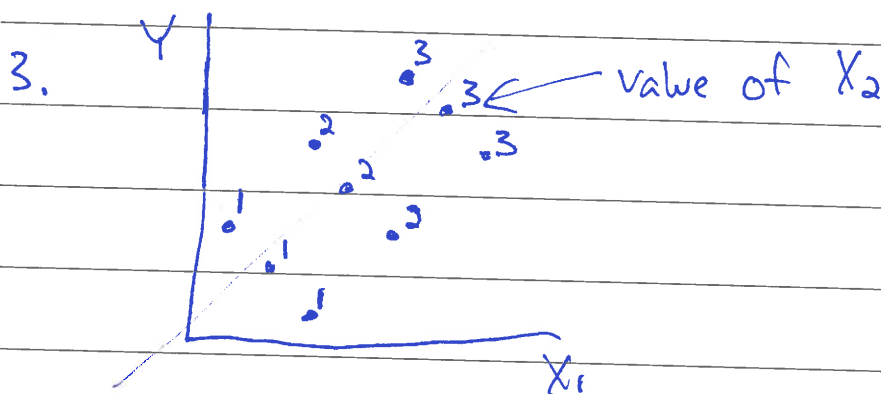
As more predictors are added to a model, there can be surprising effects on the previously-included predictors:

1. predictors can switch from being “significant” to being “not significant”
2. predictors can switch from being “not significant” to being “significant”
3. the signs on the parameter estimates can flip from negative to positive, and vice-versa

- 1 **★ Exercise:** Explain why each of the above three things could occur, in the
 2 context of going from a model with a single predictor, to a model with two
 3 predictors.

4 1. If X_2 is strongly related to X_1 (large
 5 sample correlation), then the variability
 6 in Y previously attributed to X_1 may
 7 now be attributed to X_2 and X_1 is
 8 no longer "significant"

9 2. Suppose that X_1 and X_2 are not related.
 10 X_2 explains enough of the variability in
 11 response to reduce $SE(\hat{\beta}_1)$ enough so
 12 that the t-stat is "significant."
 13



When there are two predictors (and the intercept) in a model, the least squares slope estimates can be written as

$$\hat{\beta}_1 = \left(\frac{r_{y1} - r_{y2}r_{12}}{1 - r_{12}^2} \right) \left(\frac{s_y}{s_1} \right)$$

$$r_{12} = \text{Corr}(X_1, X_2)$$

and

$$\hat{\beta}_2 = \left(\frac{r_{y2} - r_{y1}r_{12}}{1 - r_{12}^2} \right) \left(\frac{s_y}{s_2} \right)$$

where s_i is the sample standard deviation for predictor i .

★ **Exercise:** Consider three regression models:

$$Y = \beta_0 + \beta_1 x_1 + \epsilon$$

$$Y = \beta'_0 + \beta_2 x_2 + \epsilon$$

$$Y = \beta''_0 + \beta_1^* x_1 + \beta_2^* x_2 + \epsilon$$

Assume parameters are estimated using least squares. What is the relationship between $\hat{\beta}_1$ and $\hat{\beta}_1^*$ and between $\hat{\beta}_2$ and $\hat{\beta}_2^*$? (It's cleaner to restrict to the case where the response and the predictors have been rescaled to each have standard deviation of one.)